

enterprise systems, domain workflows, external services

- **RAG and embeddings:** Embedding models, vector stores for documents, KBs, FAQs, policies, retrieval pipelines and ranking logic
- **Prompt/pattern library:** Reusable prompt templates, chain/graph definitions, safety patterns and escalation prompts
- **HITL controller and coordinator:** Human-in-the-loop decision points, approval workflows, exception handling flows and routing
- **Multimodality controller:** Handling text, images, audio, and structured data interchangeably; routing to specialist models (vision, speech, code, etc)

Some of the prominent tools used in this layer are

LangChain, Toolformer, RAG pipelines, Pinecone, Weaviate, FAISS, PromptLayer, HumanLoop, Guardrails.ai.

Data and knowledge layer: This provides governed, high-quality, context-rich knowledge and data that agents can reason over. The components of this layer are:

- **Enterprise data sources:** Data warehouses, data lakes, operational DBs, domain systems
- **Knowledge repositories:** Document stores, wikis, policy repositories, SOPs, email archives, tickets, chat logs, call transcripts
- **Semantic layer:** Ontologies, taxonomies, business glossaries, knowledge graphs linking entities and relationships
- **RAG infrastructure:** Vector stores for per-domain corpora, indexing, chunking, metadata tagging
- **Data governance:** Data classification, data lineage and stewardship, access policies (by role, domain, sensitivity)

Tools used in this layer are Azure Data Lake, Snowflake, Databricks, MongoDB Atlas, Neo4j, Postgres, Elasticsearch, Milvus, Unstructured.io.

Integration layer: This layer provides robust, secure connectivity between agents, data, and enterprise applications. Various components of this layer are:

- **Enterprise APIs and microservices:** REST/GraphQL/gRPC endpoints to line-of-business systems
- **Human-in-the-loop interfaces:** Review portals, approval dashboards, case management tools, workflow inboxes
- **LLM hub and enterprise LLM gateways:** Routing across multiple models (open, closed, domain-tuned), policy-based model selection (e.g., sensitivity, latency, cost)
- **Enterprise data and knowledgebase access:** Connectors to data warehouses, data lakes, ECM systems, search
- **Event and messaging:** Pub/Sub-integration with enterprise buses (e.g., Kafka, Service Bus, Google Cloud Pub/Sub)

Tools used in this layer are Azure API Management, MuleSoft, Zapier, Retool, Temporal, Airbyte, GraphQL Federation, REST/gRPC endpoints.

Infrastructure and platform layer: This layer provides a scalable, highly available, reliable and resilient foundation for agents to execute complex tasks efficiently. The components

of this layer are:

- **Cloud platforms:** Azure, AWS, Google Cloud, NVIDIA and specialised AI clouds
- **Runtime:** GPU clusters, CPU pools, Kubernetes, serverless, container orchestrators
- **Model and LLM infrastructure:** LLM API gateways and routers, and hosted model endpoints
- **Vector and agent state databases:** Vector DBs for embeddings, NoSQL/relational stores for agent state and logs
- **Data engineering pipelines:** ETL/ELT pipelines feeding data and knowledge layer, data quality tools and schedulers
- **External services:** Identity providers, secrets management, key vaults, observability stacks

Tools used in this layer are Azure, AWS, GCP, NVIDIA DGX, Hugging Face Inference Endpoints, Kubernetes, Ray, Docker, Terraform.

Operations and observability layer: This monitors system activities, provides feedback mechanisms, and facilitates continuous improvement by optimising processes based on operational data. Its components are:

- **Performance and evaluation:** Metrics on task success, latency, cost, and user satisfaction
- **Resource and token utilisation:** Telemetry for token usage, GPU/CPU, memory, cost tracking and optimisation policies
- **Agent traces and behaviour tracking:** Full traces of tool calls, decisions, and outcomes, episode-level and step-level logs
- **Guardrails and runtime safety:** LLM safety filters, PII detection, rate limiting, policy enforcement, circuit breakers
- **Monitoring and alerts:** Dashboards for health, anomalies, drift, alerts on suspicious behavior, failure spikes, policy violations

Tools used in this layer are Arize AI, Fiddler, DeepEval, RAGAS, OpenTelemetry, Prometheus, Grafana, AgentOps, LangSmith.

Governance, security, and risk layer: This layer ensures agents operate within legal, ethical, and enterprise boundaries. Major responsibilities include:

- **Identity and access management:** Agent identities and service principles, role-based and attribute-based access control, action-level permissions
- **Policy management:** Data usage policies (PII, PHI, PCI, etc), regulatory constraints, model usage policies and provider constraints
- **Risk and compliance oversight:** Logging and auditability of every action and decision, compliance mapping (HIPAA, GDPR, SOC2, etc, as applicable)
- **Ethics and responsible AI:** Bias testing, fairness checks, human accountability and override mechanisms, incident response playbooks for AI failures

Tools used in this layer are Azure Purview, Immuta, Okta, Vault, AWS IAM, Microsoft Entra, Truera, IBM Guardium, BigID.